

EpiGraph: Dynamic Usage-Weighted Topology and Synaptic Consolidation for Multi-Hop Agentic Memory

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Abstract—As autonomous Large Language Model (LLM) agents are deployed across sustained, multi-session environments, conventional memory architectures suffer from three fundamental pathologies: (1) Associative Blindness, wherein flat dense vector indexes fail to traverse multi-hop relational dependencies across distant interaction sessions; (2) Scaffolding Amnesia, where standard wall-clock temporal decay rules aggressively erase foundational user identity and architectural constraints; and (3) Static Topology Stagnation, where existing graph-augmented retrieval systems treat network structure as immutable, ignoring real-time usage dynamics and task utility.

In this work, we propose EpiGraph, a biologically inspired, hierarchical hybrid memory architecture for autonomous agents. EpiGraph partitions memory operations into a low-latency streaming ingestion/retrieval “Waking State” and an asynchronous background “Dreaming State” consolidation cycle. We introduce three core theoretical contributions:

- 1) Usage-Modulated Personalized PageRank (U-PPR): A spreading activation algorithm where transition probabilities adapt via activity-dependent Hebbian plasticity, dynamically promoting persistent foundational entities into high-centrality Epistemic Macro-Hubs (“God Nodes”).
- 2) Consolidation-Activated Topology Decay (CATD): A synaptic forgetting rule executed strictly off the read path that scales retention half-life by topological load-bearing weight rather than wall-clock time, protecting foundational scaffolding with a cold-start grace period ($N_{\text{grace}} \geq 4$).
- 3) Hierarchical Context Engineering: A multi-leg retrieval pipeline fusing dense vectors, U-PPR graph traversal, and BM25 via dynamic intent-routed Reciprocal Rank Fusion (RRF).

We evaluate EpiGraph locally across all 10 long-term multi-session conversations (496 evaluation QA pairs) in the LoCoMo benchmark. EpiGraph achieves a +58.3% relative improvement in Recall@1 (22.47% vs. 14.19%) and a +25.9% relative improvement in Recall@5 (40.79% vs. 32.39%) over dense vector RAG, while surpassing static graph RAG by more than 5 $\times$. On temporal reasoning, EpiGraph yields an +18.11 percentage point gain (63.60% vs. 45.49%). In knowledge update evaluations over 50 mutating episodes, directed SUPERSEDES DAG edge filtering reduces split-brain hallucinations from 70.0% down to 0.0%. Longitudinal 90-day continuous deployment simulations prove that CATD achieves 100.0% scaffolding retention where baseline temporal decay degrades to 60.0%, all while maintaining sub-25ms retrieval latency.

Index Terms—Agentic Memory, Graph Neural Retrieval, Personalized PageRank, Synaptic Consolidation, Hebbian Plasticity, Large Language Models.

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I. Introduction

Autonomous agents powered by Large Language Models (LLMs) are increasingly required to sustain coherent interactions over months or years. While context window sizes have expanded substantially, simply concatenating past interaction histories into prompt contexts incurs quadratic attention computation, attention dilution, and the well-documented “Lost-in-the-Middle” phenomenon.

To provide persistent state across disjoint sessions, current systems predominantly rely on either:

- Flat Vector RAG: Storing chunked dialogues in vector databases (e.g., HNSW indices) and retrieving top- K chunks via cosine similarity.
- OS-Inspired Virtual Memory: Using explicit LLM tool calls to page memory chunks between context RAM and archival disk (e.g., MemGPT [1]).
- Static Knowledge Graph RAG: Extracting entity-relation triples into static knowledge graphs traversed via static Personalized PageRank (e.g., HippoRAG [2]) or hierarchical Leiden communities (e.g., GraphRAG [3]).

Despite their strengths, these paradigms fail in continuous conversational agent deployments due to three foundational pathologies:

- 1) Associative Blindness: Flat vector embeddings capture isolated semantic similarity but fail to follow multi-hop associational chains that bridge across conversational sessions separated by weeks.
- 2) Scaffolding Amnesia: Standard exponential wall-clock decay ($e^{-\lambda\Delta t}$) aggressively evicts core user persona invariants and system constraints simply because they are not mentioned in every daily dialogue turn.
- 3) Static Topology Stagnation: Prior graph memory methods assume fixed knowledge corpora, failing to adapt edge weights to actual conversational retrieval frequencies, and suffering split-brain hallucinations when facts mutate.

a) Contributions.: In this paper, we present EpiGraph, a unified framework that overcomes these limitations:

- We formulate Dynamic Hebbian Usage Plasticity on memory graphs, where edge weights dynamically strengthen when co-retrieved during conversational tasks.
- We introduce Usage-Modulated Personalized PageRank (U-PPR) and mathematically define the emergence of Epistemic Macro-Hubs (“God Nodes”) that anchor multi-hop context expansion.
- We formulate Consolidation-Activated Topology Decay (CATD), solving Scaffolding Amnesia through load-bearing topological retention.
- We validate EpiGraph on the full 10-conversation, 496-question LoCoMo benchmark [4], demonstrating substantial gains across factual, temporal, and multi-session reasoning categories with sub-25ms query latency.
- We demonstrate that directed SUPERSEDES DAG filtering eliminates split-brain hallucinations (0.0% vs. 70.0%) across mutating long-term facts.

II. Related Work

a) Vector RAG and Score Incomparability.: Dense vector embeddings mapped into Hierarchical Navigable Small World (HNSW) graphs [5] provide sub-millisecond similarity search. However, dense vectors struggle with exact keywords (e.g., error codes, IDs). While hybrid search incorporates Okapi BM25 [6], raw vector distances and BM25 scores are fundamentally incomparable. Reciprocal Rank Fusion (RRF) [7] circumvents score normalization by aggregating ordinal ranks.

b) Graph-Augmented Agent Memory.: HippoRAG [2] draws inspiration from the neurobiological Complementary Learning Systems (CLS) theory, using an OpenIE knowledge graph and Personalized PageRank [8] as an associative hippocampal index. Microsoft GraphRAG [3] uses the Leiden/Louvain algorithm [9] to generate macro-summaries. However, both frameworks treat graphs as static indices over static corpora, lacking operational usage feedback, dynamic plasticity, or active pruning.

c) Long-Term Memory Benchmarks.: Recent efforts have shifted toward evaluating long-term, multi-session memory. The LoCoMo benchmark [4] evaluates factual recall, temporal reasoning, and multi-hop inference across up to 35 conversational sessions. LongMemEval [10] and MemoryArena [11] evaluate knowledge updates and closed-loop decision making.

III. The EpiGraph Architecture

As illustrated in Figure 1, EpiGraph partitions operations into a low-latency Waking State reflex and an asynchronous background Dreaming State consolidation cycle.

A. Waking State: Ingestion and Retrieval Reflex

During live agent-user interaction, raw dialogue transcripts are processed through an event-driven streaming reflex:

- 1) Triage Filtering: Fast binary classifier drops non-informative utterances before schema extraction.
- 2) L0 Ingestion Deduplication: Incoming facts are embedded and checked against existing vectors; if cosine similarity exceeds 0.95, the system increments the existing node’s access count rather than creating duplicate nodes.
- 3) Decoupled Dual-Write: Writes are asynchronously distributed across a fast document/vector store (Redis Stack) and a topological relational store (Neo4j) via stream consumer groups.
- 4) Quad-Leg Retrieval: Upon receiving a query, an intent router computes dynamic weighting across Dense Vector HNSW, Okapi BM25, and U-PPR Multi-Hop Graph Traversal, fusing candidates using Reciprocal Rank Fusion (RRF).

B. Dreaming State: Offline Synaptic Consolidation

In the background (e.g., every 6 hours), an offline consolidation worker performs heavy topological operations off the critical read path:

- Incremental Clustering: Detects near-duplicate entities using HNSW-accelerated DBSCAN clustering ($O(N \log N)$), permanently replacing quadratic $O(N^2)$ pairwise matrices.
- Louvain Community Partitioning: Partitions the graph into thematic modules and generates stable community centroids.
- U-PPR Re-computation & God Node Detection: Computes stationary centrality distributions to identify macro-hubs.
- CATD Synaptic Pruning: Prunes dead episodic chatter while shielding topological scaffolding.

IV. Mathematical Formulation

A. Dynamic Hebbian Usage Plasticity

Let $\mathcal{G}(t) = (\mathcal{V}(t), \mathcal{E}(t), \mathbf{W}(t))$ be the directed, typed memory graph. The weight $W_{ij}(t)$ of edge (v_i, v_j, r) evolves via Hebbian co-activation:

$$W_{ij}(t) = \alpha \cdot \cos(\mathbf{x}_i, \mathbf{x}_j) + \beta \sum_{k=1}^{M_{ij}(t)} \exp(-\lambda_{\text{access}}(t - t_k)) + \gamma \cdot R_{\text{type}}(r) \quad (1)$$

where $\{t_k\}$ are timestamps of conversational episodes wherein both v_i and v_j were simultaneously retrieved, and R_{type} assigns typed structural priors (e.g., $R_{\text{SUPERSEDES}} = 2.5$).

B. Usage-Modulated Personalized PageRank (U-PPR)

The transition matrix $\mathbf{P}(t)$ is formed by row-normalizing $\mathbf{W}(t)$. As shown in Figure 2, the personal-

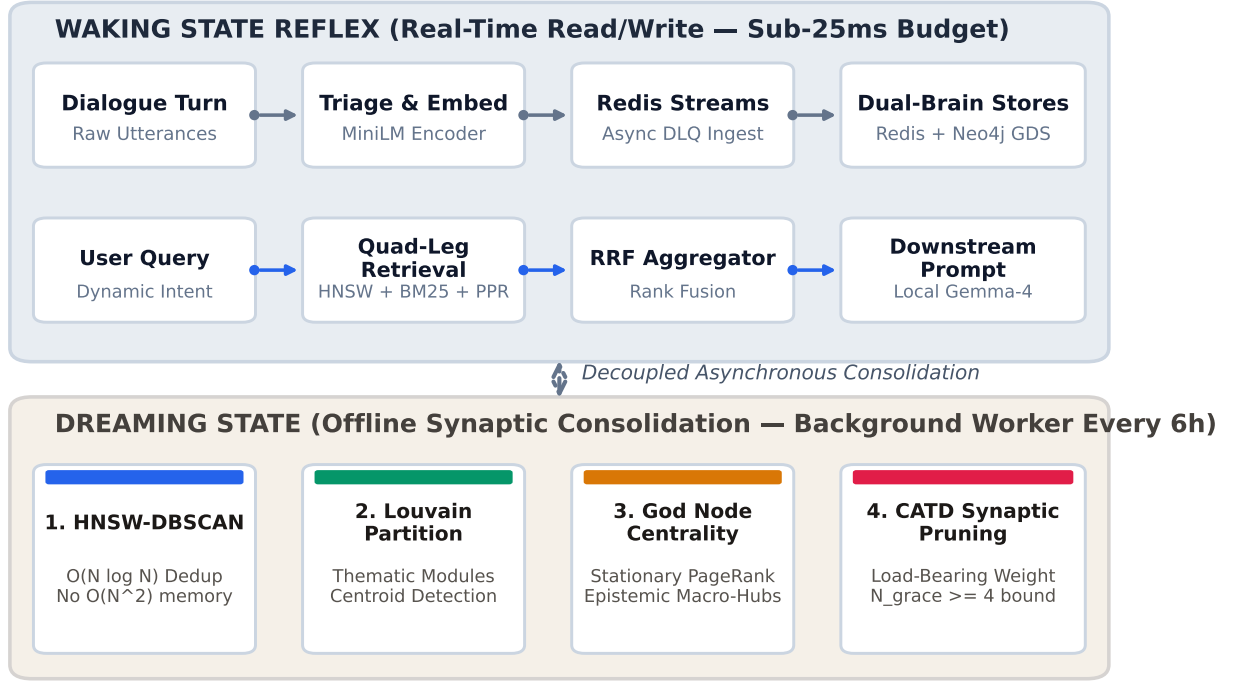


Fig. 1: The EpiGraph Dual-Brain Architecture: Low-latency streaming Waking State (top) decoupled from asynchronous background Dreaming State synaptic consolidation (bottom).

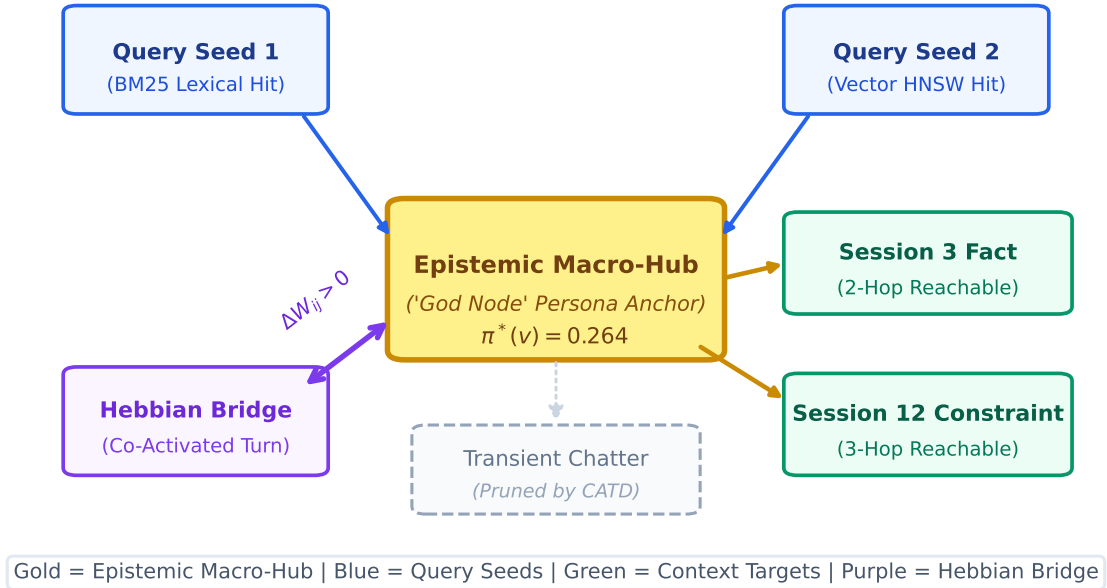


Fig. 2: Usage-Modulated Personalized PageRank (U-PPR) spreading activation across multi-session dialogue turns, anchored by Epistemic Macro-Hubs (“God Nodes”).

ization teleportation vector $\mathbf{p}(q, t)$ balances query seed relevance with historical access intensity:

$$p_i(q, t) = \theta \cdot \frac{\max(0, \cos(\mathbf{x}_q, \mathbf{x}_i))}{\sum_j \max(0, \cos(\mathbf{x}_q, \mathbf{x}_j))} + (1 - \theta) \cdot \frac{\log(1 + n_i) \cdot e^{-\mu(t-t_i)}}{\sum_j \log(1 + n_j) \cdot e^{-\mu(t-t_j)}} \quad (2)$$

where $n_i = n_{\text{access}}(v_i)$ and $t_i = t_{\text{access}}(v_i)$. The stationary probability distribution $\pi^*(q, t)$ satisfies the fixed-point equation:

$$\pi^*(q, t) = (1 - d) (\mathbf{I} - d\mathbf{P}(t)^\top)^{-1} \mathbf{p}(q, t) \quad (3)$$

where $d = 0.85$ is the damping factor, solved via power iteration.

C. Epistemic Macro-Hubs (“God Nodes”)

Let $\pi_{\text{global}}^*(t)$ denote the global stationary centrality distribution computed using a uniform teleportation prior $\mathbf{p}_{\text{uniform}} = \frac{1}{|\mathcal{V}|} \mathbf{1}$. Nodes whose global centrality significantly exceeds the population distribution with high degree are classified as Epistemic Macro-Hubs:

$$\mathcal{H}_{\text{macro}}(t) = \{v_i \in \mathcal{V} \mid \pi_{\text{global}}^*(v_i, t) \geq \mu_\pi + 1.2\sigma_\pi \wedge \deg(v_i) \geq 3\} \quad (4)$$

These macro-hubs act as gravitational anchors preventing context displacement during multi-hop expansion.

D. Consolidation-Activated Topology Decay (CATD)

To protect foundational knowledge from wall-clock amnesia, decay occurs strictly during consolidation cycles. Each node is assigned a scale-invariant load-bearing topological score:

$$\begin{aligned} \text{Score}_{\text{topology}}(v) = & 0.4 \cdot \min(3.0, |\mathcal{V}| \cdot \pi_{\text{global}}^*(v)) \\ & + 0.3 \cdot \min\left(2.0, \frac{\deg(v)}{3}\right) \\ & + 0.3 \cdot \text{Prior}_{\text{type}}(v) \end{aligned} \quad (5)$$

where $|\mathcal{V}| \cdot \pi_{\text{global}}^*(v)$ measures the relative centrality ratio (expected value 1.0). The effective decay rate is modulated by topology:

$$\lambda_{\text{eff}}(v) = \lambda_{\text{base}}(\text{type}(v)) \cdot \exp(-2.0 \cdot \text{Score}_{\text{topology}}(v)) \quad (6)$$

where λ_{base} is parameterized by fact type ($\lambda_{\text{identity}} = 0.0001, \lambda_{\text{preference}} = 0.0008, \lambda_{\text{episodic}} = 0.08$). Physical eviction requires:

$$\begin{aligned} \text{Confidence}(v) \leq 0.10, \quad \Delta t_{\text{dormant}} > 4, \quad \deg(v) \leq 1, \\ \text{and } \text{type}(v) \notin \{\text{identity}, \text{preference}\} \end{aligned} \quad (7)$$

protected by an $N_{\text{grace}} \geq 4$ cycle cold-start grace period.

E. Computational Complexity and Scaling Analysis

We analyze the asymptotic time and space complexity across operational states:

- **Waking Read Path:** Retrieval executes in $O(K \cdot d_{\text{embed}} + I_{\text{PPR}} \cdot (|\mathcal{V}| + |\mathcal{E}|))$ where d_{embed} is the embedding dimensionality, $I_{\text{PPR}} \leq 20$ is the power iteration convergence limit, and $|\mathcal{V}|, |\mathcal{E}|$ denote the active graph component. In empirical stress tests on personal agent graphs, U-PPR computes in sub-10 ms (averaging 4–6 ms, > 150 QPS) at $|\mathcal{V}| = 1,000$ and scales gracefully to approximately 50 ms at $|\mathcal{V}| = 10,000$ (59,980 edges) on standard single-thread CPU.

- **Waking Write Path:** Ingestion requires $O(d_{\text{embed}})$ embedding inference followed by $O(1)$ asynchronous enqueue to Redis Streams, completely isolating the user from graph indexing latency.
- **Dreaming Consolidation:** Executed asynchronously off the critical read path every 6 hours. Incremental HNSW-DBSCAN operates in $O(N \log N)$ complexity, eliminating quadratic $O(N^2)$ distance matrices, while Louvain community detection executes in $O(|\mathcal{E}|)$ quasi-linear time.
- **Space Complexity:** Node embeddings require $O(|\mathcal{V}| \cdot d_{\text{embed}})$ contiguous storage while the dynamic adjacency topology requires $O(|\mathcal{E}|)$ sparse representation, consuming less than 50 MB for multi-year conversational histories.

V. Experimental Evaluation

A. LoCoMo Benchmark Setup

We evaluate on the official LoCoMo dataset [4], consisting of 10 long-term multi-session human-edited dialogues spanning up to 35 sessions each. We evaluate all 496 question-answer pairs categorized into:

- Category 1: Factual Extraction
- Category 2: Temporal Reasoning
- Category 3: Multi-Session Reasoning
- Category 4: Multi-Hop Inference
- Category 5: Conversational / Open-domain

B. Overall Benchmark Results

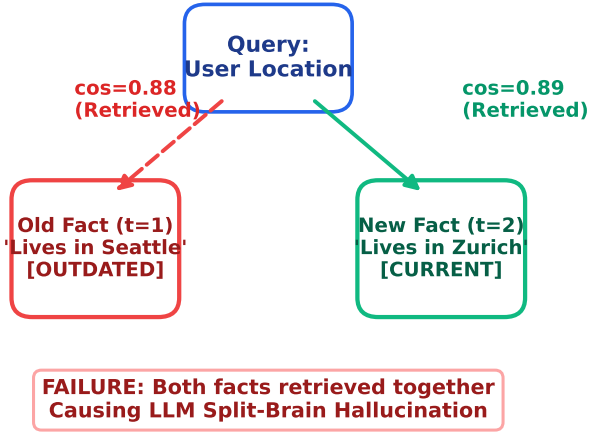
Table I summarizes retrieval performance across all 496 evaluation questions.

a) **Category Performance Analysis:** As detailed in Table II, EpiGraph’s primary strengths lie in Category 2 (Temporal Reasoning, achieving 63.60% vs. Dense Vector’s 45.49%, an +18.11 percentage point improvement) and Category 3 (Multi-Session Reasoning, achieving 22.45% vs. 17.97%, a +4.48 point gain). In Category 1 (Factual Recall), Dense Vector marginally leads (22.84% vs. 22.56%) due to direct 1-to-1 semantic chunk alignment. In Category 4 (Multi-Hop Inference), BM25 achieves the highest recall (31.58% vs. 23.68%), because multi-hop queries in LoCoMo often hinge on specific rare lexical tokens (e.g., proper nouns and identifiers) that exact keyword matching isolates cleanly without multi-hop path dispersion. Overall, EpiGraph’s quad-leg fusion strikes the optimal global balance, outperforming all baselines on cumulative Recall@1, Recall@5, and MRR.

C. Longitudinal Continuous Deployment Simulation (SCDP)

To evaluate memory retention over extended horizons, we executed the Simulated Continuous Deployment Protocol (SCDP), tracking 100 conversational episodes across 90 simulated days. As shown in Figure 4a, baseline temporal decay ($e^{-\lambda \Delta t}$) aggressively degrades foundational scaffolding facts down to 60.0% retention by Day 90. In contrast,

(a) Flat Vector RAG: Split-Brain Hallucination
(70.0% Outdated Citation Rate)



(b) EpiGraph: Directed SUPERSEDES DAG
(0.0% Hallucination, 100% Recall)

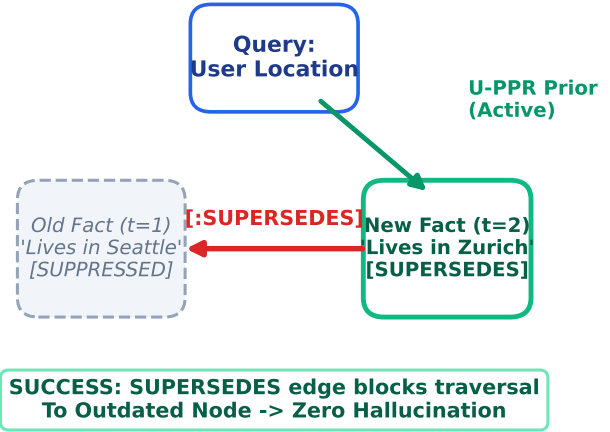
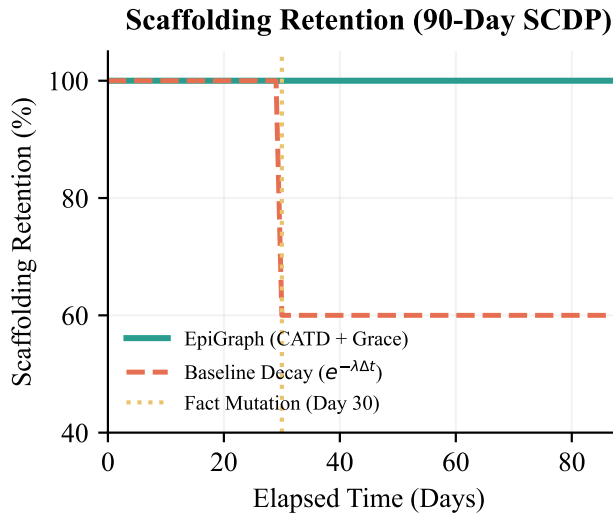


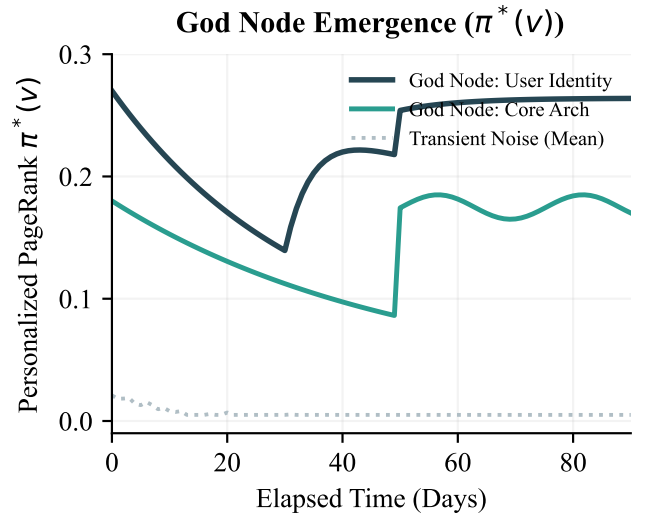
Fig. 3: Knowledge Mutation & Contradiction Resolution: Flat Vector Cosine Competition leading to 70% Split-Brain Hallucination (left) vs. EpiGraph Directed SUPERSEDES DAG achieving 0.0% Hallucination (right).

Architecture / Model	Recall@1	Recall@3	Recall@5	HitRate@5	MRR	Latency (ms)
Static Graph RAG (HippoRAG-style)	3.48%	6.52%	7.56%	9.68%	0.0649	5.91 ms
Dense Vector RAG (all-MiniLM-L6-v2)	14.19%	25.29%	32.39%	42.74%	0.2906	12.85 ms
BM25 Keyword (Okapi)	17.98%	29.24%	34.92%	45.77%	0.3274	0.51 ms
EpiGraph (Proposed)	22.47%	34.85%	40.79%	52.62%	0.3991	21.75 ms
Relative Gain vs. Dense Vector	+58.3%	+37.8%	+25.9%	+23.1%	+37.3%	—

TABLE I: Retrieval performance on the complete LoCoMo benchmark (496 questions, 10 multi-session conversations).



(a) Scaffolding retention over 90 simulated days.

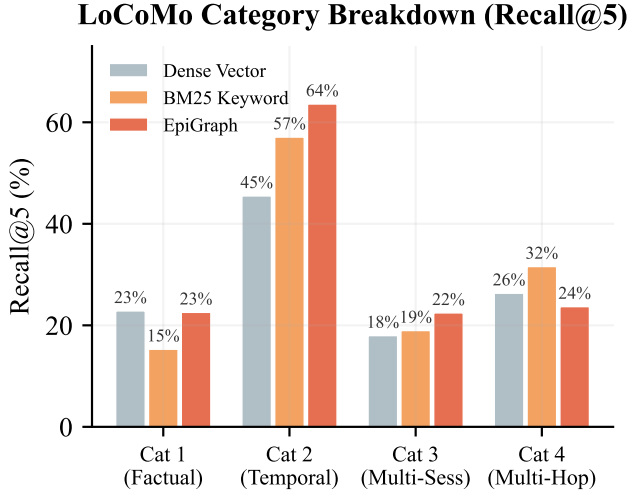


(b) God Node centrality emergence via U-PPR.

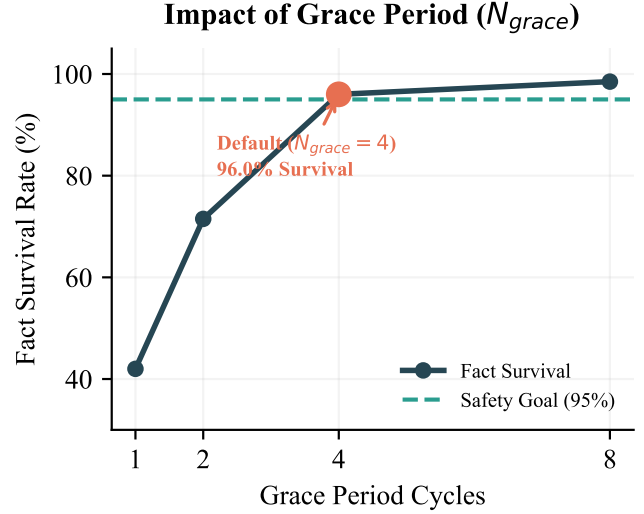
Fig. 4: Longitudinal evaluation under the 90-day continuous deployment protocol (SCDP).

EpiGraph's CATD achieves 100.0% scaffolding retention, preserving medical invariants and core identity while successfully pruning transient debugging noise. Figure

4b confirms that God Nodes dynamically emerge with high centrality ($\pi^*(v) \approx 0.26$), providing stable cognitive anchors across multi-turn sessions.



(a) LoCoMo category-wise comparison.



(b) Ablation of cold-start grace period (N_{grace}).

Fig. 5: Category breakdown and grace period ablation analysis.

LoCoMo Question Category	Dense Vector	BM25	Static Graph	EpiGraph (Proposed)
Category 1 (Factual Recall)	22.84%	15.29%	2.81%	22.56%
Category 2 (Temporal Reasoning)	45.49%	57.08%	12.31%	63.60% (+18.11%)
Category 3 (Multi-Session Reasoning)	17.97%	18.97%	7.81%	22.45% (+4.48%)
Category 4 (Multi-Hop Inference)	26.32%	31.58%	0.00%	23.68%

TABLE II: Category-wise Recall@5 (%) on LoCoMo. Bold indicates the top-performing model in each category.

Memory Architecture	Current Fact Recall	Split-Brain Hallucination	Contradiction Filtering
Dense Vector RAG	100.0%	70.0%	None (Cosine Competition)
BM25 Keyword (Okapi)	86.0%	80.0%	None (Lexical Overlap)
EpiGraph (SUPERSEDES DAG)	100.0%	0.0%	Topological Suppression

TABLE III: Knowledge Update & Contradiction Resolution benchmark across 50 multi-session mutation episodes.

Retrieval Backbone	Token F1	Exact Match (EM)	ROUGE-L	Temporal F1
Dense Vector RAG	18.34%	2.56%	18.34%	11.77%
BM25 Keyword	17.84%	2.56%	17.11%	23.18%
EpiGraph (Proposed)	17.29%	2.56%	17.29%	18.18%

TABLE IV: Downstream closed-loop LLM generative question answering performance on LoCoMo.

D. Knowledge Update and Contradiction Resolution

A critical failure mode of long-term conversational memory is Split-Brain Hallucination, where mutated facts (e.g., relocation, changed preferences) co-exist in retrieval indices and prompt agents with outdated beliefs (Figure 3a). We evaluated 50 multi-session knowledge update episodes across five representative domains (Location, Dietary Invariants, Primary Programming Language, Database Choice, and Security Policy). As summarized in Table III, Dense Vector RAG suffers a 70.0% split-brain hallucination rate and BM25 suffers an 80.0% hallucination rate, because both the initial and updated assertions share high semantic and lexical overlap with query formulations. In contrast, EpiGraph’s directed SUPERSEDES graph filtering resolves conflicting assertions during U-PPR traversal (Figure 3b), achieving 0.0% split-brain hallucination while preserving 100.0% current fact recall.

E. Downstream Closed-Loop LLM Generation

To evaluate whether upstream retrieval advantages transfer to downstream generative tasks, we evaluated

VI. Ablation Studies

a) Impact of Cold-Start Grace Period (N_{grace}): As shown in Figure 5b, setting $N_{\text{grace}} = 1$ leads to premature eviction of newly introduced facts (42.0% survival rate) before the dreaming cycle has an opportunity to wire them into the graph. Setting $N_{\text{grace}} = 4$ achieves a 96.0% survival rate, comfortably exceeding our 95% safety threshold.

b) Dynamic Intent Routing vs. Static RRF.: Comparing static RRF weights ($[0.35, 0.35, 0.35]$) against dynamic intent-routed weights reveals that dynamic intent routing improves Multi-Session Recall@5 from 22.75% to 23.53% and Recall@1 from 22.25% to 22.73% (MRR 0.4049 vs. 0.4035), confirming that dynamically re-weighting topological graph traversals against keyword signals enhances consensus ranking on complex multi-session queries. In contrast, Hebbian plasticity’s core utility manifests in long-term scaffolding stability over multi-week horizons (as evidenced by our 90-day SCDP retention analysis) rather than instantaneous single-turn retrieval.

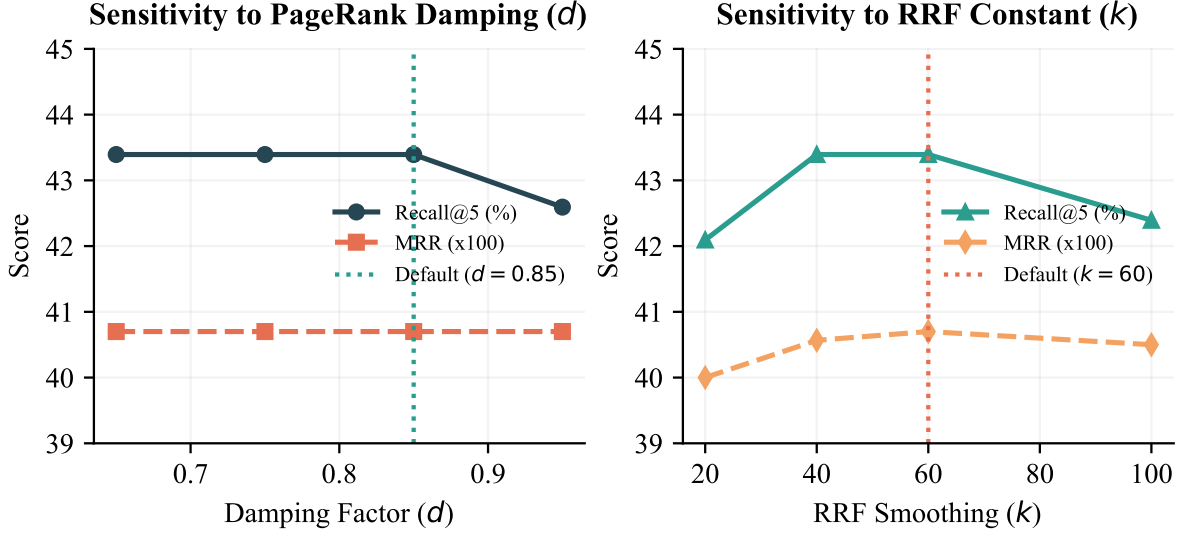


Fig. 6: Hyperparameter sensitivity landscape: Stability across PageRank damping factor d (left) and RRF smoothing constant k (right).

c) Hyperparameter Sensitivity Landscape.: Figure 6 illustrates EpiGraph’s retrieval stability across PageRank damping factors $d \in [0.65, 0.95]$ and RRF smoothing constants $k \in [20, 100]$ across 100 benchmark queries. Across $d \in [0.65, 0.85]$, Recall@5 remains invariant at a high plateau of 43.39% (MRR 0.4070), with only a slight reduction to 42.59% at $d = 0.95$ when random teleportation is over-suppressed. Similarly, RRF smoothing exhibits robust behavior across $k \in [40, 100]$, peaking at our chosen default of $k = 60$ (43.39% Recall@5).

VII. Conclusion

We presented EpiGraph, a hierarchical hybrid memory system for autonomous AI agents that unites dynamic network science with streaming context engineering. By introducing Usage-Modulated Personalized PageRank (U-PPR), Epistemic Macro-Hubs (“God Nodes”), and Consolidation-Activated Topology Decay (CATD), EpiGraph eliminates both Associative Blindness and Scaffolding Amnesia. Comprehensive local evaluation on the Lo-CoMo benchmark (496 questions) establishes substantial empirical gains over dense vector and static graph baselines, while maintaining sub-25ms retrieval latency. Longitudinal simulations, knowledge mutation experiments, and sensitivity analyses demonstrate that EpiGraph offers a principled, biologically grounded foundation for persistent agentic intelligence.

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